



Lackey Clinic

A Free and Charitable Healthcare Center

From Patient Churning to Machine Learning

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What is Lackey?

Lackey Clinic is a free and **low-cost health clinic** in Yorktown, Virginia, dedicated to providing **preventative and long-term care** for uninsured or underinsured patients. Their mission is to **keep patients healthy when they cannot afford it!**

What's the problem?

The clinic faces **high patient churn** (44% of patients churn) and does not know why this is happening. This **disrupts treatment plans, leads to worse health outcomes, and creates extra administrative costs** because onboarding new patients is time-intensive and highly regulated.



How did we solve this?

Using patient demographics, diagnoses, visit history and much more, we built **machine-learning models (KNN and Logistic Regression)** to understand who is **most likely to churn**. This allowed us to identify **key patterns** behind disengagement and **create a tool** that can **flag at-risk patients early**.

What we found!

Churn was more common among patients with **unstable housing**, those who are **unemployed or self-employed**, patients **without chronic conditions**, and **older adults who become eligible for Medicaid**. **Our model can accurately identify 84 out of 100 churners**, helping the clinic accurately intervene sooner.



What should Lackey Do?

Lackey can reduce churn by **emphasizing its telehealth services** to patients with irregular schedules, **emphasizing the importance of routine check-ins** for those who come in for acute issues, and **connecting patients with housing or support services** when needed. **Continuing strong language services** is also crucial, as non-English-speaking patients tend to remain highly engaged with the clinic.

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